

Markov Modeling of Evolutionary Algorithms

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Abstract

Evolutionary Algorithms (EAs) are population-based optimization methods, widely applied in engineering, machine learning, operations research, and related fields. Accurately modeling the dynamics of these algorithms remains a challenge. While traditional Markov models have been used to analyze mutation-only EAs, these approaches typically fail when crossover is introduced, due to its dependence on pairs of parents. In this work, we present a Markov model that incorporates both crossover and mutation without tracking individual genotypes. The key contribution of our approach is the construction of a crossover-mutation transition tensor, which encodes the probability of producing offspring of a given fitness from two parents with specific fitness levels. Combined with tournament selection and blind replacement, this tensor defines a Markov process over population fitness distributions. We validate the model by comparing its predicted fitness trajectories against empirical EA simulations, demonstrating close agreement with actual dynamics. This framework offers a computationally efficient tool for understanding EA behavior and supports further exploration of evolutionary dynamics.

1. Background and Motivation

Evolutionary Algorithms (EAs) are population-based metaheuristics inspired by the principles of natural selection and genetic variation. At each generation, a population of individuals evolves through the stochastic operators of **selection**, **variation**, and **replacement**, driven by a **fitness function** that rewards high-performing individuals.

The theoretical modeling of EAs has received significant attention. Most models focus on mutation-only settings, where a single parent produces an offspring via bit-flip mutation. In such models, population dynamics can often be captured via simple Markov chains. However, these models largely omit **crossover**, which is central to the performance of many EAs.

Modeling crossover introduces two key difficulties. First, it involves two parents, making the offspring distribution dependent on *pairs of individuals*, which breaks the independence assumption of standard Markov chain models. Second, the *interaction between fitness-based selection and variation* leads to dynamics that are difficult to analyze in closed form.

In this work, we propose such a framework by introducing a **population-level distribution model** that operates directly on the fitness distribution of the population rather than on individual genomes. The central feature of our approach is the construction of a **third-order crossover and mutation transition tensor**, which encodes the probability of producing an offspring with a given fitness from two parents of specified fitness levels. This tensor captures the behavior of the variation operators in isolation.

By combining this transition tensor with a tournament selection-based parent distribution and assuming blind replacement (no elitism), we obtain a deterministic **Markov process over population fitness distributions**. This approach provides a structured, analytically feasible model of EA dynamics with crossover, which we validate against empirical simulations.

Our goal is to bridge the gap between low-level simulation and high-level analytical modeling, providing a tool that captures key features of EA behavior while simplifying the complexity of direct simulation. Although we focus here on tournament selection and the OneMax fitness function without elitism, the framework we present offers a foundation for further exploration.

2. Framework and Abstraction

2.1. Genotype Space

Let $n \in \mathbb{N}$ be a fixed genome length, and $\mathbf{G} = \{0, 1\}^n$ be the set of all bitstrings of length n . Each genome $x = (x_1, x_2, \dots, x_n) \in \mathbf{G}$ represents a candidate solution in the search space. The total number of genotypes is $|\mathbf{G}| = 2^n$, which quickly becomes infeasible for analysis. We therefore exploit the structure of the fitness function to abstract the space.

2.2. Fitness Function

We define the fitness function as the OneMax function:

$$f(x) = \sum_{i=1}^n x_i$$

which simply counts the number of 1s in the genome of x . The fitness values range from 0 to n , so the fitness space is:

$$\mathbf{F} = \{0, 1, 2, \dots, n\}$$

2.3. Population Distribution over Fitness Levels

Rather than tracking the full distribution of genotypes, we consider the **fitness distribution** of a population. Let $v^{(t)} = [v^{(t)}[0], v^{(t)}[1], \dots, v^{(t)}[n]]$ denote the fitness distribution of the population at generation t , where

$$v^{(t)}[j] = \mathbb{P}[f(x) = j], \forall j \in \mathbf{F}$$

Here $v^{(t)}$ is a standard probability vector, so $v[j] \geq 0, \sum_{j=0}^n v[j] = 1$.

By focusing on this fitness distribution, we **abstract away genotype-level variation** and reduce the dimensionality of the problem from exponential in n to linear in $n + 1$. This allows for tractable analysis while retaining the key information needed to model evolutionary dynamics.

3. Tournament Selection and Parent Distribution

We define the parent selection strategy used in our model: **tournament selection**. This is a standard selection method in evolutionary algorithms, widely used due to its simplicity and tunable selection pressure. Our focus is on modeling how tournament selection transforms a fitness distribution $v^{(t)}$ into a parent-pair distribution.

3.1. Single Parent Selection Distribution

Recall $v^{(t)} = [v^{(t)}[0], v^{(t)}[1], \dots, v^{(t)}[n]]$ is the fitness distribution of the population at generation t . Fix a **tournament size** $k \geq 1$. To sample a single parent under tournament selection, we perform the following:

1. Sample k individuals from the population with replacement, each sampled according to the current population distribution $v^{(t)}$. This is a group in the tournament.
2. Select the individual among that group with the highest fitness. This is the "winner" of the tournament.

Let $\rho^{(t)}[i]$ denote the probability that a genome with fitness level $i \in \mathbf{F}$ is selected as a parent under this procedure. Then:

$$\rho^{(t)}[i] = \left(\sum_{j=0}^i v^{(t)}[j] \right)^k - \left(\sum_{j=0}^{i-1} v^{(t)}[j] \right)^k$$

This expression represents the probability that the maximum fitness among k individuals is exactly i .

3.2. Parent Pair Distribution

We assume that the two parents involved in crossover are selected independently, each through the same tournament selection procedure.

Let $\rho^{(t)}[i_1][i_2]$ denote the probability that a mating pair is selected with fitness levels i_1 and i_2 , respectively. Then, under the independence assumption,

$$\rho^{(t)}[i_1][i_2] = \rho^{(t)}[i_1] \cdot \rho^{(t)}[i_2]$$

This parent-pair distribution determines how likely any given fitness combination is to contribute to the next generation. It serves as the input to the variation process, which we model through the crossover and mutation transition tensor introduced in the next section.

4. Crossover Transition Tensor

Having defined how parents are selected through tournament selection, we now model the process by which two parents generate offspring through crossover and mutation. To capture this, we introduce a third-order transition tensor that encodes the probabilities of offspring fitness outcomes given the fitness levels of both parents.

4.1. Definition of the Tensor

We define a third-order tensor $P \in [0, 1]^{(n+1) \times (n+1) \times (n+1)}$, where:

$$P[i_1][i_2][j] = \mathbb{P}(\text{offspring has fitness } j \mid \text{parents have fitness } i_1 \text{ and } i_2)$$

This quantity represents the probability that two parents with fitness levels i_1 and i_2 produce an offspring of fitness j , through crossover and mutation operations.

Each slice $P[i_1][i_2]$ is a valid probability distribution, so $\sum_{j=0}^n P[i_1][i_2][j] = 1$, $\forall i_1, i_2, \in \mathbf{F}$.

This tensor captures only the behavior of the variation operator, independent of the current population distribution or selection mechanism.

4.2. Estimation via Simulation

Since it is typically infeasible to derive P in closed form, we estimate it empirically through simulation. For each parent pair (i_1, i_2) , the procedure is as follows:

1. Generate two parent genomes $x_1, x_2 \in \{0, 1\}^n$ such that $f(x_1) = i_1$, $f(x_2) = i_2$.
2. Apply a crossover operator (one-point crossover) to produce an intermediate offspring.
3. Apply a mutation operator (bit-flip) to each bit of the offspring with probability $p_m \in [0, 1]$. Let y be the resulting offspring genome.
4. Record the fitness of the offspring $j = f(y)$.
5. Repeat the above steps for a large number of trials, and normalize the observed frequencies to obtain $P[i_1][i_2][j]$ for each $j \in \mathbf{F}$.

4.3. Interpretation

The transition tensor P provides a complete characterization of the variation operators at the level of fitness classes. For every pair of parent fitness levels (i_1, i_2) , the entry $P[i_1][i_2][j]$ represents the probability that their offspring will have fitness j . Thus, this tensor precisely encapsulates how variation transforms fitness distributions, independent of selection dynamics or population context.

Empirically estimating this tensor through simulations of individual parent-pair interactions is a necessary and justified methodological choice. Direct analytical derivation of P is typically infeasible due to the complexity introduced by crossover and mutation operations at the genotype level. However, once estimated, this tensor remains fixed and independent of the ongoing evolutionary process. This means the estimation is conducted strictly as a preliminary step, entirely separate from any subsequent population-level analyses.

In other words, although empirical simulation is used to build P , the Markov model itself does not rely on iterative simulation of individual genomes or any ongoing empirical measurements. The tensor functions as a reusable, precomputed component, providing a general abstraction of variation that can be used flexibly with any selection method or population distribution.

This separation between the empirical estimation of the tensor and the application of the Markov model clearly distinguishes between the localized behavior of genetic operators and global population dynamics.

5. Offspring Distribution and Markov Process

With the parent selection and variation operators fully defined and encoded, we now describe how the population evolves from generation t to $t + 1$. Specifically, we derive the offspring distribution from the parent-pair distribution $\rho^{(t)}$ and the variation transition tensor P , and use this to define the Markov process governing population dynamics.

5.1. Offspring Distribution

The expected probability that an offspring has fitness $j \in \mathbf{F}$ at generation $t + 1$ is obtained by summing over all possible parent pairs (i_1, i_2) , weighted by their selection probabilities $\rho^{(t)}[i_1][i_2]$ and the variation probabilities $P[i_1][i_2][j]$.

Formally, the offspring distribution $o^{(t)} = [o^{(t)}[0], o^{(t)}[1], \dots, o^{(t)}[n]]$ is given by:

$$o^{(t)}[j] = \sum_{i_1=0}^n \sum_{i_2=0}^n \rho^{(t)}[i_1][i_2] \cdot P[i_1][i_2][j]$$

This distribution represents the expected proportion of offspring with each fitness level, given the current generation's fitness distribution and the defined operators.

5.2. Blind Replacement

Our model assumes **blind replacement**, where the population at generation $t + 1$ consists entirely of offspring generated according to $o^{(t)}$. No individuals from the current generation are guaranteed to carry over, so there is no form of elitism.

Under this assumption, the population fitness distribution at generation $t + 1$ is given by:

$$v^{(t+1)} = o^{(t)}$$

5.3. Markov Process

The update rule defined above induces a deterministic, discrete-time Markov process on the space of fitness distribution.

$$v^{(t+1)} = \mathcal{M}(v^{(t)})$$

This process satisfies the Markov property because:

- The parent pair distribution $\rho^{(t)}$ is computed entirely from the current fitness distribution through tournament selection.
- The offspring distribution is generated from $\rho^{(t)}$ via the fixed crossover tensor P , independent of any past generations.
- The replacement strategy is blind, meaning the next generation is fully determined by the offspring distribution alone.

The operator $\mathcal{M}$ in our model is **nonlinear**, due to the nonlinearity introduced by tournament selection in determining parent-pair distributions. Therefore, standard linear Markov chain matrices cannot be directly used to represent the evolutionary transitions. Despite the nonlinearity, the model’s Markov property provides a structured framework for studying population dynamics.

6. Model Validation

To evaluate the accuracy of the described Markov chain model, we compared its predicted fitness dynamics against the results of direct evolutionary algorithm simulations.

Both the simulation and the Markov process were implemented in Python, and all code used for this study is available [here](#).

6.1. Experimental Setup

The goal of the experiment was to compare the mean fitness trajectory predicted by the Markov model with the average outcome of many independent EA simulation runs. Both the model and the simulation were run under identical evolutionary parameters:

Parameter	Value
Population size	500
Genome length	10
Number of generations	30
Mutation rate	0.05 (bit-flip per bit)
Tournament size	2
Fitness function	OneMax (sum of 1s)
Replacement strategy	Blind generational replacement
Variation operators	One-point crossover, followed by mutation

The validation consisted of two separate procedures:

1. **EA Simulation:** The simulation tracked explicit genomes (bitstrings), applying tournament selection, crossover, mutation, and blind replacement at each generation. The process was repeated for 100 independent runs, each initialized with a random population. The average fitness over generations was computed by averaging across these runs.
2. **Markov Model:** The model operated on the fitness distribution of the population. The initial fitness distribution was computed from a randomly generated population (matching the EA simulation setup). A precomputed third-order crossover and mutation tensor

$P[i_1][i_2][j]$ was constructed over 100 trials for each parent fitness pair. The model then updated the fitness distribution over 30 generations using this tensor and tournament-based parent selection. As in the simulation, the average fitness over generations was computed.

6.2. Results

Figure 1 shows the average fitness across generations from the EA simulation and the expected fitness computed by the Markov model.

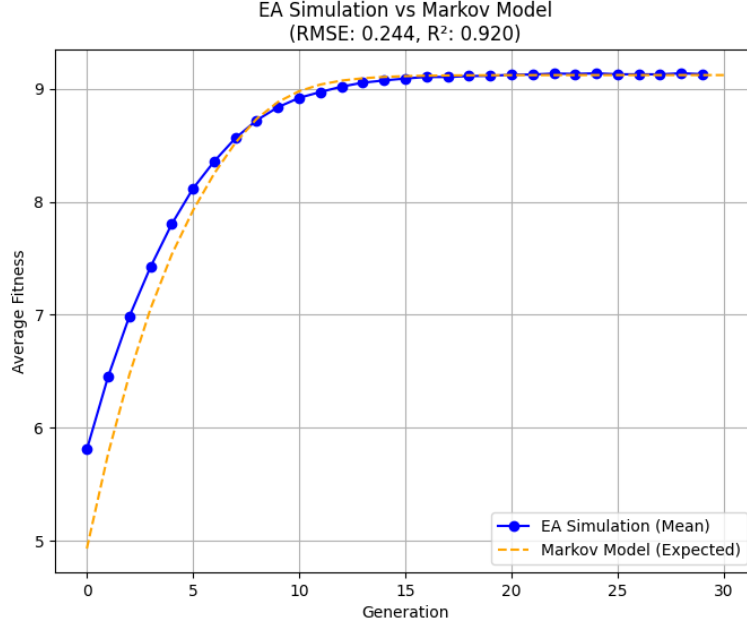


Figure 1: Average fitness from EA simulations (blue, solid) and predicted average fitness from Markov model (orange, dashed).

The RMSE of 0.244 and R^2 of 0.920 demonstrate that the Markov chain model provides an effective approximation of the EA behavior, closely capturing the overall trajectory of the fitness evolution observed in the simulations.

7. Discussion

The central goal of this paper was to develop an accurate model of evolutionary algorithms (EAs) that incorporates the effects of crossover and mutation without explicitly simulating individual genomes. To achieve this, we proposed a framework that abstracts away individual genotypes and instead models the population at the level of fitness distributions. This abstraction dramatically reduces the complexity of the state space. Instead of tracking 2^n genotypes, the model operates on just $n + 1$ fitness levels, significantly reducing the complexity of analysis.

While general Markov models are standard in mutation-only settings, they fail when crossover is introduced, because crossover inherently depends on pairs of individuals. A standard Markov chain approach assumes that the next state depends only on the current state, but with crossover, the offspring distribution depends on joint parent interactions, breaking this independence. A key feature of this work is the construction of a crossover and mutation transition tensor. The tensor explicitly encodes the probability distribution of offspring fitness levels given the fitness levels of two parents. This construction enables the use of a Markov process at the fitness distribution level, even when crossover is present.

By combining the fitness distribution abstraction with the estimated crossover-mutation tensor, we provide a framework that remains analytically tractable while accurately capturing the behavior of EAs.

Limitations and Further Research

Through experimental validation, we found that the proposed Markov model accurately captured the overall dynamics of fitness improvement across generations. However, the current framework was evaluated under the following assumptions:

- No elitism (only blind replacement)
- Only tournament selection with fixed tournament size
- Only OneMax fitness function

Future work could explore these extensions and examine how well the model predicts additional EA metrics, such as convergence time and diversity maintenance.

Ultimately, this work contributes to a broader effort to build analytical tools for understanding evolutionary algorithms. By bridging simulation and theory, this framework offers a step toward deeper analytical understanding of evolutionary algorithm dynamics and their underlying mechanisms.